

COURSE PLAN

Odd/Even Semester/2025-26

SECTION I

Semester	5th	Programme & Class	MSc AI/ML
Course Code	MAI 573B	Course Type	3L+4P (Full CIA)-LCA
Course Title	Quantum Computing	LCA	Yes
Total Hours	75	Hours per week	07
Faculty Name	Dr. Somnath Sinha	Contact Details and Office Hours	somnath.sinha@christuniversity.in +91-9717620165
Class Policies and Guidelines	Guidelines for classes: • Strictly adhere to the class policies of the University/Department • Active and voluntary participation is expected during class hours. • Unbiased peer support is encouraged. • Usage of the Laptop is only with the prior permission (as per the requirements of the class) As per the time slot given in the timetable, students are expected to be seated in their respective classrooms.		
Course Description	This course covers the basics of quantum computing, including complex numbers, linear algebra, and quantum mechanics postulates, along with practical skills in designing quantum gates, circuits, and error correction. Students will learn about quantum algorithms like Deutsch's and Grover's, explore quantum information theory, and understand pragmatic applications such as cryptography and Quantum Key Distribution for secure communication.		

Course Objectives (Optional)	The course aims to provide learners with foundational knowledge of quantum computing, including quantum states, circuits, algorithms, machine learning, and cryptography. It seeks to develop analytical, coding, and problem-solving skills, enabling students to design, simulate, and apply quantum techniques for research, innovation, and secure computational applications.			
Course Outcomes	• CO1: Understand foundational knowledge of quantum computing, including its fundamental principles, algorithms, and potential applications. • CO2: Develop proficiency in modeling quantum circuits using various quantum computation environments and frameworks, enabling the implementation and simulation of quantum algorithms. • CO3: Acquire a deep understanding of quantum operations, including noise and error-correction techniques, and their significance in mitigating errors and enhancing the reliability of quantum computations. • CO4: Explore various quantum algorithms such as Deutsch’s algorithm, the Deutsch–Jozsa algorithm, the Quantum Fourier transform, Grover’s Algorithm, and understand their applications in solving computational problems exponentially faster than classical algorithms			
SECTION II				
Week	Module/ Unit/ Topic details	Andragogy	Ungraded formative Assessments	Resource/ Reference details
Week 01 (11/09/2025 to 13/09/2025)	Complex Numbers – Linear Algebra - Matrices and Operators - Global Perspectives Lab Exercise 1 (Implementation of basic quantum operations using matrices and operators)	Need to Know	Diagnostic Quiz	Textbook 1: Chapter 1 & 2
Week 02 (15/09/2025 to 19/09/2025)	Postulates of Quantum Mechanics – Quantum Bits - Representations of Qubits -Superpositions	Learners’ Experience	Peer Assessment	Textbook 1: Chapter 3

	Lab Exercise 2 (Simulating quantum gates and their effects)			
Week 03-04 (22/09/2025 to 04/10/2025) CIA I	Universal logic gates - Basic single qubit gates - Multiple qubit gates - Lab Exercise 3 (Development of quantum circuits to perform specific tasks or algorithms)	Self-Directed Learning	Think-Pair-Share (During Theory)	Textbook 1: Chapter 8
Week 05 - 06(06/10/2025 to 18/10/2025) CIA II	Circuit development - Quantum error correction Lab Exercise 4 (Understanding gate composition and circuit optimization)	Learners' Experience	Pre-Lab Quiz	Textbook 1-Chapter 8 &17
Week 07 (27/10/2025 to 31/10/2025)	Quantum parallelism - Deutsch's algorithm - The Deutsch-Jozsa algorithm - Quantum Lab Exercise 5 (Implementation and testing of quantum parallelism concepts)	Problem-Based Learning	Concept maps	Textbook 1: Chapter 10
Week 08 (03/11/2025 to 08/11/2025)	Fourier transform and its applications - Quantum Search Algorithms: Grover's Algorithm Lab Exercise 6 (Demonstrating the parallel processing capabilities of quantum	Need to Know	Think-Pair-Share	Textbook 1: Chapter 10

	algorithms)			
Week 09 (10/11/2025 to 14/11/2025)	Data compression - Shannon's noiseless channel coding theorem Lab Exercise 7 (Simulating various error scenarios)	Relevance (Problem- Centered)	Checkpoint Activities (During Lab)	Textbook 1: Chapter 5
Week 10 (17/11/2025 to 22/11/2025) CIA III	Schumacher's quantum noiseless channel coding theorem - Classical information over noisy quantum channels Lab Exercise 8 (Evaluating error- correction techniques)	Learners' Experience	Peer Feedback	Textbook 1: Chapter 5
Week 11-12 (24/11/2025 to 06/12/2025)	Classical cryptography basic concepts - Private key cryptography Lab Exercise 9 (Implementing encryption algorithms)	Relevance (Problem- Centered)	Mini-Project Extension	Textbook 1: Chapter 11 & 12
Week 13 (08/12/2025 to 13/12/2025)	Shor's Factoring Algorithm - Quantum Key Distribution - BB84 - Ekert 91 Lab Exercise 10 (Simulating Shor's Factoring Algorithm, and testing Quantum Key Distribution protocols)	Relevance (Problem- Centered)	Demo and Peer review	Textbook 1: Chapter 7

Week 14 (16/12/2025 to 22/12/2025) ETE				
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Textbooks:

1. William Chuck, Easttom II, Quantum computing fundamentals. Addison-Wesley Professional, 2021.
2. Parag K Lala, Mc Graw Hill Education, “Quantum Computing, A Beginners Introduction”, First edition,2020.

Essential Reading / Recommended Reading:

1. Scott Aaronson, “Quantum Computing Since Democritus”, Cambridge University Press, 2013.
2. N. David Mermin, “Quantum Computer Science: An Introduction”, Cambridge University Press, 2007.

SECTION III

Note: Refer to the syllabus entry in Espro (Except for CO-PO strength mapping, rest of the details will be populated based on your Espro Syllabus entry)

Mapping:

Programme Outcomes	CO1	CO2	CO3	CO4
PO1	2	2	3	2
PO2	2	2	2	2
PO3	2	3	2	3
PO4	1	1	2	1
PO5	2	2	2	2

NOTE: Please add /delete rows and columns where necessary to accommodate all the Programme Outcomes and Programme Specific Outcomes.

Mapping of needs with the syllabus:

Course Outcomes	Local, Regional, National, Global and Development Needs (LRNG)	Cross-cutting issues relating to Professional Ethics, Gender, Human Values, Environment and Sustainability	Employability/Entrepreneurship/ Skill Development	Sustainable Development Goals
CO1	Foundational knowledge of quantum computing addresses local to global needs by enabling research, innovation, and technology development for societal progress.	Quantum computing education promotes ethics, inclusivity, human values, and sustainability in advanced technology applications.	It equips learners with future-ready skills, fostering employability, entrepreneurship, and interdisciplinary expertise.	Quantum computing contributes to quality education, innovation, climate action, and global partnerships for sustainable growth. (SDG-4,9,13)
CO2	Proficiency in modeling quantum circuits addresses local to global needs by supporting research, technological advancement, and innovation in quantum computing.	Learning to model quantum circuits fosters ethical practice, inclusivity, responsibility, and sustainable use of emerging technologies.	It enhances employability and entrepreneurial opportunities by developing advanced skills in quantum frameworks and algorithm implementation.	Circuit modeling proficiency contributes to quality education, innovation-driven industry, sustainable solutions, and global collaborations. (SDG-4,9,13)
CO3	Understanding quantum operations and error-correction	It promotes ethical responsibility, inclusivity, and sustainability in	Learners gain specialized skills in error mitigation and reliability,	Quantum error-correction knowledge advances quality education, resilient

	meets local to global needs by enabling reliable computing for research, industry, and societal applications.	developing trustworthy quantum technologies.	strengthening employability and innovation potential.	infrastructure, sustainable innovation, and global partnerships. (SDG-4,9,13)
CO4	Exploring quantum algorithms addresses local to global needs by advancing computational research, innovation, and problem-solving capabilities.	It encourages ethical use, inclusivity, and sustainability in applying powerful quantum algorithms to real-world challenges.	Learners build strong analytical and algorithmic skills, enhancing employability and entrepreneurial opportunities in quantum technologies.	Mastery of quantum algorithms supports quality education, fosters innovation-driven industry, contributes to sustainable solutions, and strengthens global partnerships. (SDG-4,9,13)
Topic/Module Details				
I	Learning the basics of quantum computing fulfills local to global needs by building foundational capacity for research, innovation, and technological growth.	It promotes ethical awareness, inclusivity, and sustainability while introducing learners to responsible use of emerging technologies.	Foundational knowledge equips learners with essential skills that enhance employability and entrepreneurial readiness in quantum-related fields.	Understanding basic quantum concepts supports quality education, innovation, sustainable development, and international collaboration. (SDG-4,9,13,17)
II	Quantum circuit design addresses local to global needs by enabling	It cultivates ethical responsibility, inclusivity, and sustainability in the design and	Learners acquire practical skills in circuit modeling and optimization, enhancing	Circuit design knowledge supports quality education, fosters industry innovation,

	innovation in computing systems, secure communication, and advanced technologies.	application of quantum systems.	employability and entrepreneurial potential.	contributes to sustainable solutions, and encourages global partnerships. (SDG-4,9,13,17)
III	Quantum algorithms meet local to global needs by providing advanced computational methods that accelerate innovation and problem-solving.	They encourage ethical use, inclusivity, and sustainability in applying quantum power to real-world challenges.	Mastery of algorithms builds strong analytical, programming, and research skills that enhance employability and entrepreneurial opportunities.	Quantum algorithm knowledge advances quality education, drives innovation and infrastructure, supports sustainable solutions, and promotes global collaborations. (SDG-4,9,13,17)
IV	Quantum machine learning addresses local to global needs by enabling breakthroughs in data analysis, optimization, and intelligent systems.	It promotes ethical AI practices, inclusivity, and sustainability in applying quantum-enhanced learning models.	Learners gain cutting-edge skills in quantum AI, boosting employability and opening entrepreneurial pathways in emerging tech industries.	Quantum machine learning supports quality education, fosters innovation, aids climate and sustainability modeling, and strengthens global research partnerships. (SDG-4,9,13,17)
V	Quantum cryptography fulfills local to global needs by ensuring secure communication systems for research, industry, governance, and	It strengthens ethical responsibility, inclusivity, and sustainability in developing trustworthy and privacy-preserving technologies.	Learners acquire specialized skills in secure communication and cryptographic protocols, enhancing employability and entrepreneurial opportunities.	Quantum cryptography contributes to quality education, fosters resilient infrastructure, advances sustainable digital security, and supports global partnerships.

	society.			(SDG-4,9,13,17)
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A template to map the Course Outcomes against the components of the assessment is given below:

Course Outcomes	Components of assessment (with the breakup of marks)				
	CIA-I (25 M)	CIA-II (50 M)	CIA-III (25 M)	CIA -IV (25 M)	ETE (50 M)
CO 1	13	15	5	5	10
CO 2	12	20	10	5	10
CO 3	-	15	10	5	15
CO4	-	-	-	10	15

Note: Add more rows if there are more COs

Assessment outline:

	Components of assessment This will be populated from the syllabus				
Components	CIA I	CIA II	CIA III	CIA IV	ESE
Marks/Percentage	25 M	50 M	25 M	25 M	50 M

SECTION IV

Assessment Description: CIA-I

Individual examination based on syllabus given to the entire class and feedback from students/teacher

Assessment Component	Practical Examination
Assignment Topic	Basic of Quantum Measurements
Units Cover	Unit 1
Nature of the assignment	Individual Upload (presentation, documentation)
Submission mode	Individual
Maximum marks	25
Assignment Description	Performance measuring Rubrics 1) Coding (5 M) 2) Content (5 Marks) 3) Design and Presentation (5 Marks) 4) Understanding (5 Marks) 5) Viva (5 Marks)

General Instruction	● Use clear headings, diagrams, and flowcharts wherever necessary. ● Questions will be based on the submitted work as well as fundamental concepts of quantum measurements.
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Learning Outcomes:

- LO1: Understand the fundamental principle of quantum computing
- LO2: Increasing presentation skill and discussion ability.

Mapping the Learning Outcomes of the assignment with components of the evaluation rubrics:

Learning Outcomes of the assignment	Method of assessment	Component of the evaluation rubrics
LO1: Understand the fundamental principle of quantum computing	Demonstration of Understanding	Clarity in concepts
LO2: Increasing presentation skill and discussion ability.		

Assignment description: CIA-II

Theoretical test will be conducted during regular class hours

Assessment Component	Mid Trimester Examination
Descriptive test portion	Unit 1, Unit 2 & 3
Mode of Conduct	Offline
Date of Examination	
Type of questions	
Maximum marks	50 Marks
Duration	120 minutes
General Instruction	

Learning Outcomes:

LO1: Understand the fundamental principle of quantum computing
LO2: Design of quantum circuits and simulating quantum algorithm.
LO3: Understanding quantum cryptography

Evaluation Rubrics:

Score	Impression
40-50	Proficient
25-40	Satisfactory
01-25	Need to improve

Mapping the Learning Outcomes of the assignment with components of the evaluation rubrics:

Learning Outcomes of the assignment	Method of assessment	Component of the evaluation rubrics
LO1: Understand the fundamental principle of quantum computing	Offline Test	Impression
LO2: Design of quantum circuits and simulating quantum algorithm.		
LO3: Understanding quantum cryptography		

Assessment Description: CIA-III

Individual examination based on syllabus given to the entire class and feedback from students/teacher

Assessment Component	Practical Examination
Assignment Topic	Quantum Machine Learning
Units Cover	Unit 2 & 3
Nature of the assignment	Individual Upload (presentation, documentation)

Submission mode	Individual
Maximum marks	25
Assignment Description	Performance measuring Rubrics 6) Coding (5 M) 7) Content (5 Marks) 8) Design and Presentation (5 Marks) 9) Understanding (5 Marks) 10) Viva (5 Marks)
General Instruction	● Use clear headings, diagrams, and flowcharts wherever necessary. ● Questions will be based on the submitted work as well as fundamental concepts of quantum measurements.

Learning outcomes

LO1: Design of quantum circuits and simulating quantum algorithm.

LO2: Understanding quantum machine learning

Mapping the Learning Outcomes of the assignment with components of the evaluation rubrics:

Learning Outcomes of the assignment	Method of assessment	Component of the evaluation rubrics
LO 1: Design of quantum circuits and simulating quantum algorithm.	Lab Test	Impression
LO 2: Understanding quantum machine learning		

